

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I V-RUPASHREE (192319012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Rupa

85

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Student 28: RUPASHREE V | Register Number: 192565088

1. A vacuum-cleaner robot operates in a two-room environment (Room A and Room B), each of which can be clean or dirty. Define the PEAS (Performance measure, Environment, Actuators, Sensors) description for this agent, and characterize its task environment along the dimensions of observability, determinism, episodicity, and number of agents.
2. A ride-hailing company operates a network of pickup points connected by roads with travel costs (time + fuel). Using the Uniform Cost Search (UCS) algorithm, determine the least-cost path from a starting point S to a destination G. Describe the frontier and explored set at each step for a graph of your choice with at least 6 nodes.
3. Explain Iterative Deepening Search (IDS) and trace its execution on a graph of your choice with at least 6 nodes, showing the search at depth limits 0, 1, and 2. Discuss why IDS combines the space-efficiency of DFS with the completeness of BFS.

Student 29: Ravi Teja Y V | Register Number: 192319012

1. Formulate the 8-puzzle problem (a 3x3 sliding tile puzzle) as a search problem by specifying the state representation, initial state, goal test, and the set of possible actions. Explain why the branching factor of this problem is at most 4.
2. A hospital wants to build an AI agent to triage incoming patients based on symptom severity. Define an appropriate performance measure for this agent and explain the concept of rationality in this context — under what conditions would the agent's action be considered rational?
3. A warehouse company wants to deploy robots to pick and transport goods to packing stations. Identify which agent architecture (simple reflex, model-based, goal-based, utility-based, or learning agent) is most appropriate, and justify your choice with reasons specific to the warehouse environment.

Student 30: SANJANAA SREE P | Register Number: 192311454

1. An e-commerce company wants to deploy a delivery drone to autonomously deliver packages within a city. Identify what type of problem-solving agent is suitable for this scenario, and write pseudocode outlining how the agent formulates and solves the delivery routing problem.
2. Explain Iterative Deepening Search (IDS) and trace its execution on a graph of your choice with at least 6 nodes, showing the search at depth limits 0, 1, and 2. Discuss why IDS combines the space-efficiency of DFS with the completeness of BFS.
3. Solve the Missionaries and Cannibals problem: 3 missionaries and 3 cannibals must cross a river using a boat that can carry at most 2 people, such that cannibals never outnumber missionaries on either bank. Formulate the problem as a search problem (state, initial state, actions, transition model, goal test) and show the first three states explored using Breadth-First Search.

2. Formulate the 8-puzzle problem (a 3x3 sliding tile puzzle) as a search problem by specifying the state representation, initial state, goal test, and the set of possible actions. Explain why the branching factor of this problem is at most 4.
3. For a Tic-Tac-Toe playing agent, define its PEAS description and characterize the environment (is it fully observable? deterministic? episodic? single or multi-agent?). Explain why this environment is classified as adversarial.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Vacuum Cleaner Robot - PEAS & TASK Environment Analysis

PEAS

- Maximize cleanliness of Room A & Room B.
- Minimize energy consumption.
- Minimize cleaning time.
- Avoid unnecessary movements.

Environment :-

- Two rooms: Room A & Room B.
- Each room can be either Clean or Dirty.

Actuators :-

- Move left
- Move right
- Suck
- No Operation

Sensors :-

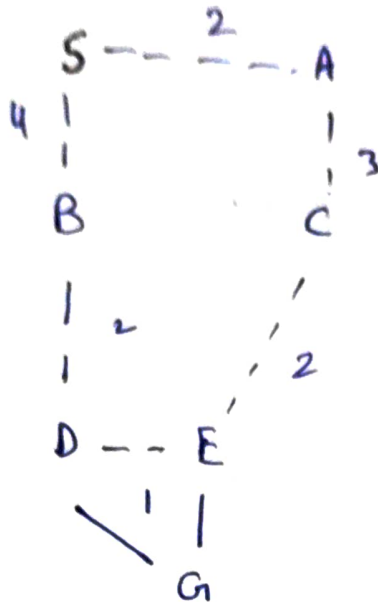
- Dirt sensor
- Location sensor

TASK

The vacuum cleaner operates in a partially observable, deterministic, sequential and single-agent environment.

2) Uniform Cost Search (UCS)

Graphs



$$S \rightarrow B = 4$$

$$B \rightarrow D = 2$$

$$D \rightarrow G = 1$$

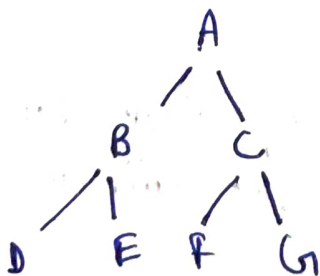
$$\text{Total Cost} = 4 + 2 + 1 = 7$$

Result:-

$$S \rightarrow B \rightarrow D \rightarrow G$$

$$\text{Total cost} = 7$$

3) Iterative Deeping Search (IDS)



Depth limit = 0

Visited nodes

A

Goal not found

Depth limit = 1

Visited nodes

A → B → C

Goal Not found.

Depth limit = 2

A → B → D → E → C → F → G

Goal Found

Solution Path:-

A → C → G

